

Control Systems II Achievement Test 2

2026.6.24

Q1. Modeling and Transfer Function (15pt)

An LR circuit is described by the following differential equation:

$$L \frac{di(t)}{dt} + Ri(t) = v(t),$$

where $i(t)$ is the current and $v(t)$ is the input voltage. Also,

$$L = 2 \text{ H}, R = 4\Omega.$$

Assume zero initial condition.

(1) Derive the transfer function from the input voltage $v(t)$ to the current $i(t)$.

$$I(s) = \mathcal{L}\{i(t)\}, V(s) = \mathcal{L}\{v(t)\}$$

The differential equation can be transformed to

$$L(sI(s) - i(0)) + RI(s) = V(s)$$

Since $i(0) = 0$,

$$(Ls + R)I(s) = V(s)$$

$$\text{Then. } \frac{I(s)}{V(s)} = \frac{1}{Ls + R} = \frac{1}{2s + 4}$$

(2) Determine the pole of the system.

$$\text{The denominator } Ls + R \rightarrow 0 \Rightarrow s = -\frac{R}{L} = -\frac{4}{2} = -2$$

(3) Determine whether the system is stable or unstable and briefly justify your answer.

Stable since the pole is a negative real number

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Q2. Time Response (15pt)

Using the transfer function obtained in Q1, suppose that the input is a unit-step signal.

$$v(t) = u(t),$$

where the Laplace transform of $u(t)$ is $1/s$.

(1) Derive the Laplace transform of the current, $I(s)$.

$$I(s) = G(s) \cdot U(s) = \frac{1}{2s+R} \cdot \frac{1}{s} = \frac{1}{2s^2+4s}$$

(2) Derive the time response of the current $i(t)$.

$$\begin{aligned} i(t) &= \mathcal{L}^{-1}\{I(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{2s+R} \cdot \frac{1}{s}\right\} \\ &= \mathcal{L}^{-1}\left\{\left(\frac{1}{s} - \frac{L}{2s+R}\right)\frac{1}{R}\right\} \\ &= \frac{1}{R} \left(1 - e^{-\frac{R}{L}t}\right) \\ &= \frac{1}{4} (1 - e^{-2t}) \end{aligned}$$

(3) Determine the steady-state value of the current and briefly justify your answer.

$$\lim_{t \rightarrow \infty} i(t) = \lim_{t \rightarrow \infty} \frac{1}{R} \left(1 - e^{-\frac{R}{L}t}\right) = \frac{1}{R} = \frac{1}{4}$$

Q3. Bode Diagram (25 points)

The following table shows the measured frequency response of an unknown linear system.

The frequency response was measured at angular frequencies of 1, 2, and 5 times each power of ten. Since the gain and phase changed rapidly in a certain frequency range, additional measurements were taken at intermediate frequencies.

ω [rad/s]	Gain	Phase [deg]	Gain [dB]
0.1	1	0	0
0.2	1	-1	0
0.5	1.02	-2	0
1	1.06	-5	0.8
2	1.31	-11	2.3
3	2.03	-27	6
4	3.33	-90	10.4
5	1.48	-146	3.5
6	0.75	-160	-1.9
10	0.19	-172	-14
20	0.042	-176	-28
50	0.0064	-179	-40

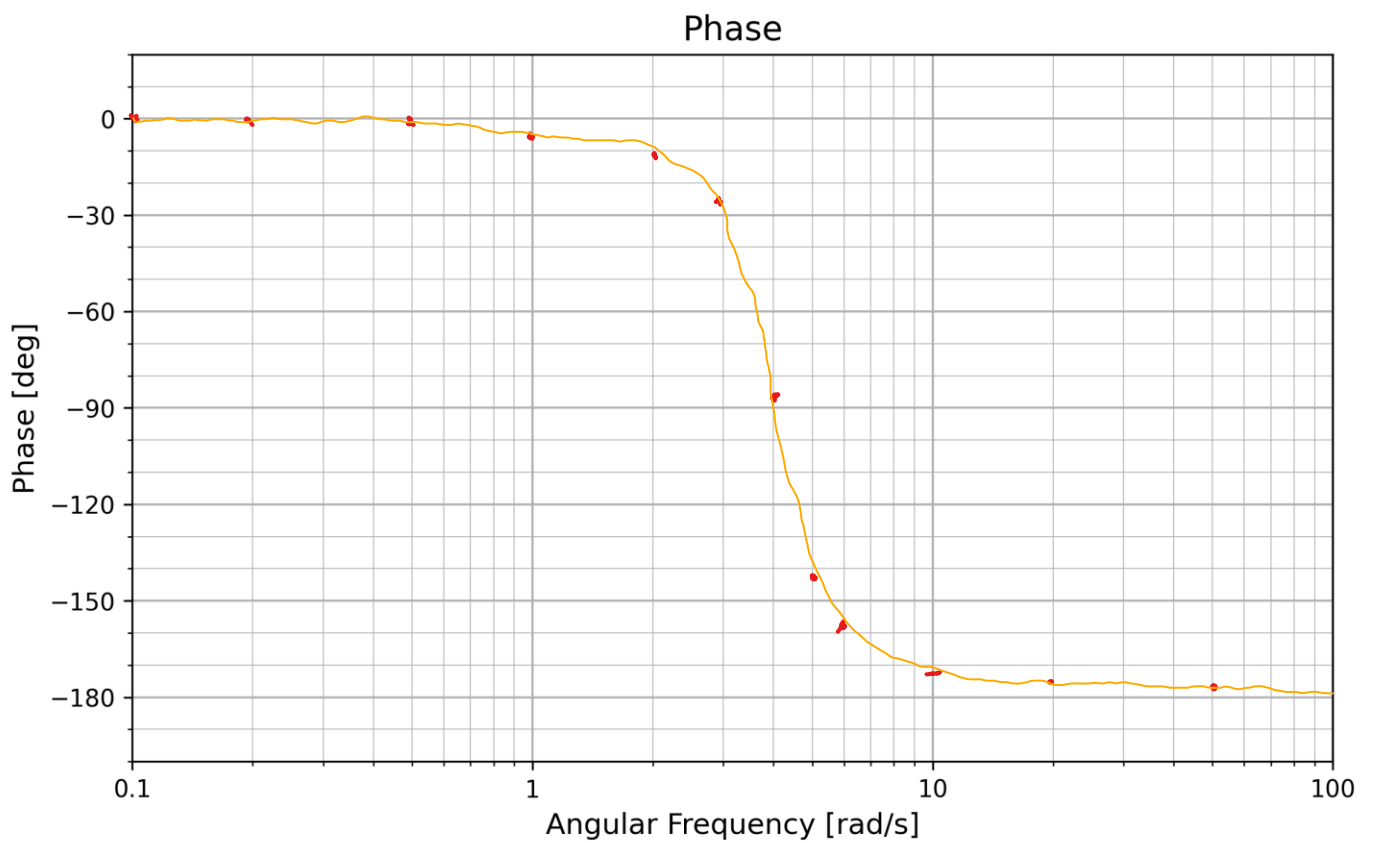
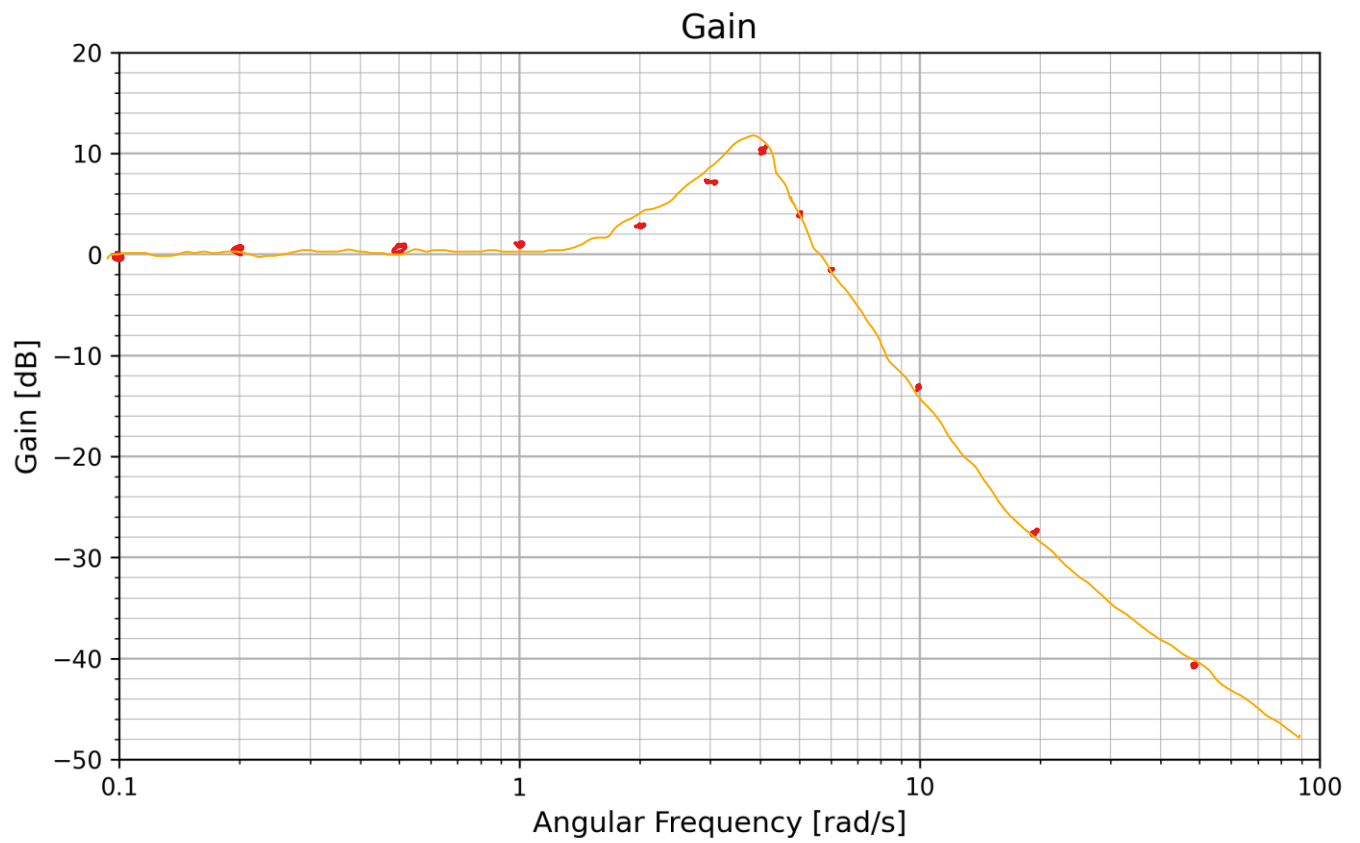
- (1) Convert the gain values into decibels using the provided $20\log x$ table and fill in the blank cells in the table above.
- (2) Plot the gain on the semi-log graph paper for gain.
- (3) Plot the phase on the semi-log graph paper for phase.
- (4) Does the system exhibit resonance? If so, at approximately what angular frequency?

True. Around 4 rad/sec

- (5) Explain why the gain becomes maximum around this frequency in terms of the locations of the poles.

The transfer function has the pole whose imaginary part corresponding to this frequency.

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Table

Gain	$20\log_{10}(\text{Gain})$ [dB]	Gain	$20\log_{10}(\text{Gain})$ [dB]
0.01	-40.0	2.6	8.3
0.02	-34.0	2.7	8.6
0.03	-30.5	2.8	8.9
0.04	-28.0	2.9	9.2
0.05	-26.0	3	9.5
0.06	-24.4	3.1	9.8
0.07	-23.1	3.2	10.1
0.08	-21.9	3.3	10.4
0.09	-20.9	3.4	10.6
0.1	-20.0	3.5	10.9
0.2	-14.0	3.6	11.1
0.3	-10.5	3.7	11.4
0.4	-8.0	3.8	11.6
0.5	-6.0	3.9	11.8
0.6	-4.4	4	12
0.7	-3.1	4.1	12.3
0.8	-1.9	4.2	12.5
0.9	-0.9	4.3	12.7
1	0	4.4	12.9
1.1	0.8	4.5	13.1
1.2	1.6	4.6	13.3
1.3	2.3	4.7	13.4
1.4	2.9	4.8	13.6
1.5	3.5	4.9	13.8
1.6	4.1	5	14
1.7	4.6	5.5	14.8
1.8	5.1	6	15.6
1.9	5.6	6.5	16.3
2	6	7	16.9
2.1	6.4	7.5	17.5
2.2	6.8	8	18.1
2.3	7.2	8.5	18.6
2.4	7.6	9	19.1
2.5	8	9.5	19.6
		10	20

Q4. Nyquist Diagram (25 points)

The following table shows the same frequency response as in Q3 in terms of its real and imaginary parts.

ω [rad/s]	Real Part	Imaginary Part
0.1	1.001	-0.008
0.2	1.002	-0.015
0.5	1.014	-0.039
1	1.06	-0.085
2	1.282	-0.256
3	1.808	-0.930
4	0	-3.333
5	-1.231	-0.821
6	-0.708	-0.255
10	-0.187	-0.027
20	-0.042	-0.003
50	-0.006	-0.000

- (1) Plot the values of the open-loop transfer function of the system on the attached complex plane.
- (2) Sketch the Nyquist diagram by connecting the plotted points with a smooth curve.
- (3) Draw the trajectory for negative frequencies.
- (4) Indicate the direction of increasing angular frequency.
- (5) Determine whether the point $(-1,0)$ is encircled by the Nyquist diagram.

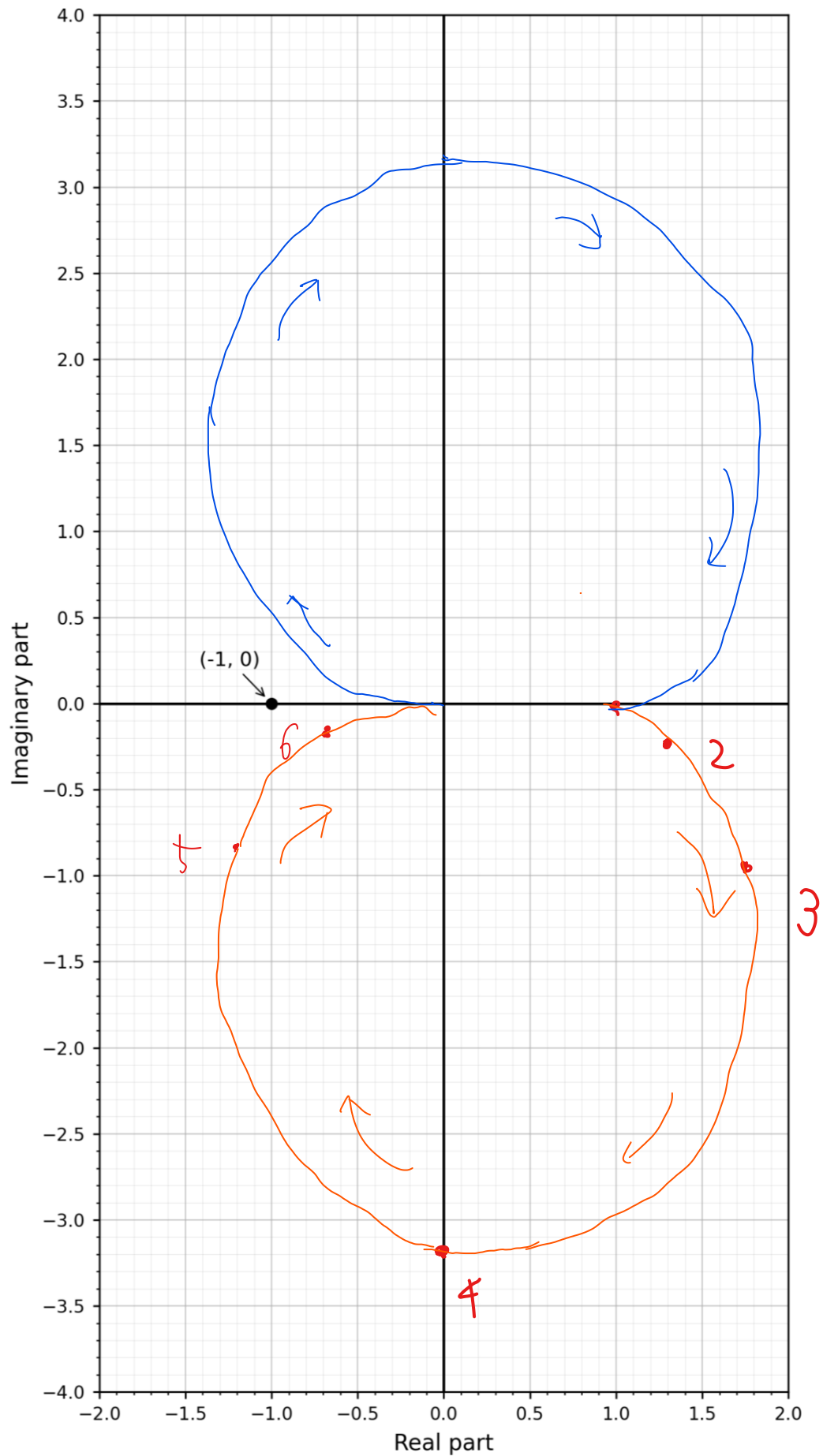
False (as plotted)

- (6) Assuming that the open-loop system has no right-half-plane poles, determine whether the corresponding closed-loop system is stable and briefly justify your answer.

Stable because the Nyquist plot does not encircle $(-1, 0)$

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Complex Plane for Nyquist Diagram



Q5. Controller Design (20 points)

The attached figure shows four closed-loop step responses labeled A–D. Responses A and B were obtained using the same type of controller with different parameter settings.

- (1) Identify the controller used for each response and explain your reasoning. Identify which responses correspond to P, PI, and PID control.
- (2) Briefly justify your answers. In particular, explain why responses A and B are different although they correspond to the same controller. Refer to the rise time, steady-state error, overshoot, and damping characteristics.

(A) Controller: P

Reason:

Steady-state error $\rightarrow$ I not used
Slow convergence $\rightarrow$ Small P gain.

(B) Controller: P

Reason:

Steady-state error $\rightarrow$ I not used
Fast convergence $\rightarrow$ Large P gain

(C) Controller: PI

Reason:

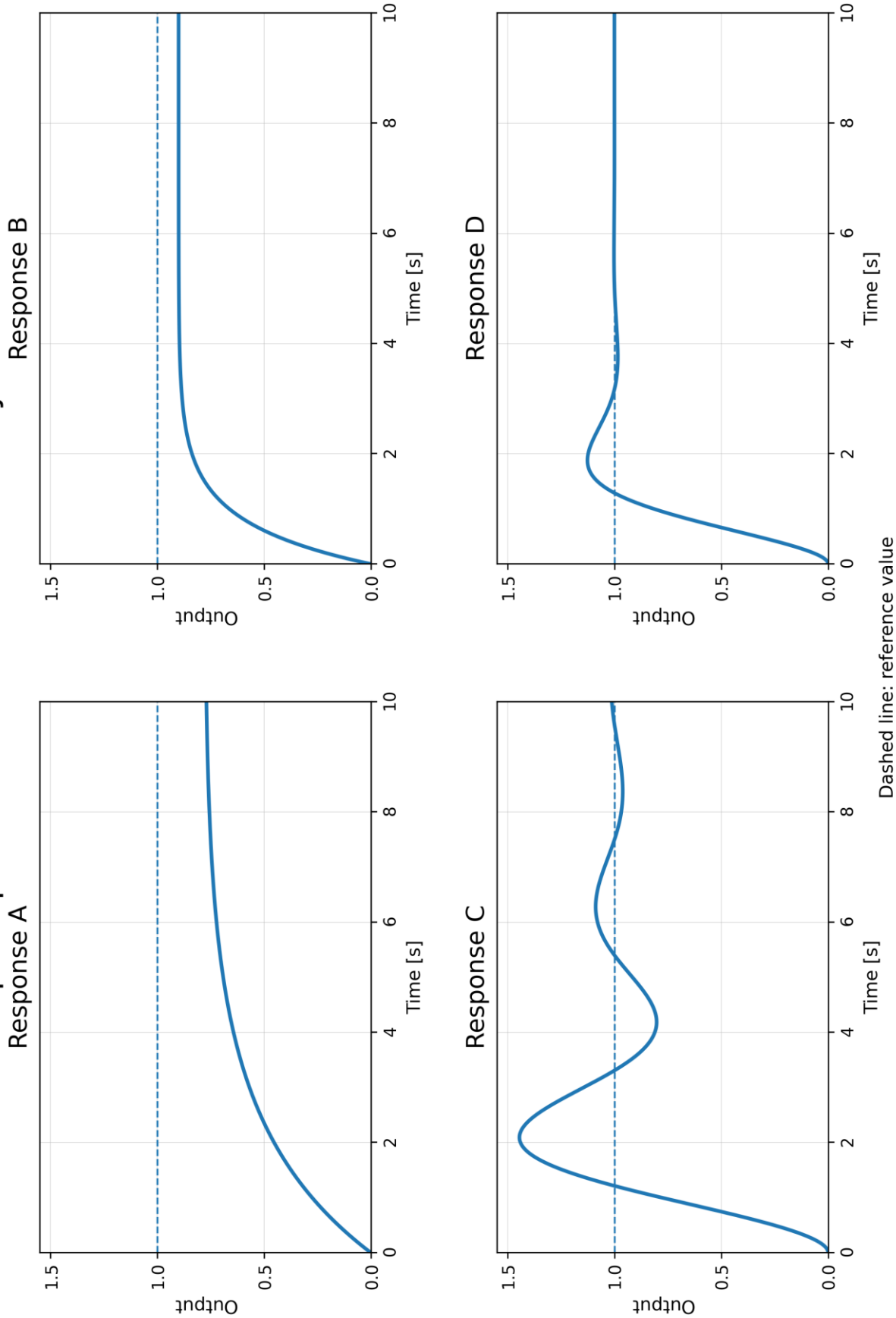
No steady-state error $\rightarrow$ I used
But oscillatory response $\rightarrow$ D not used

(D) Controller: PID

Reason:

No steady-state error $\rightarrow$ I used.
No oscillation, small overshoot $\rightarrow$ D used.

Step Responses of Feedback Control Systems



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$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t	$\frac{1}{s^2}$
$t^n \ (n = 1, 2, \dots)$	$\frac{n!}{s^{n+1}}$
e^{-at}	$\frac{1}{s+a}$
$1 - e^{-at}$	$\frac{a}{s(s+a)}$
$\sin at$	$\frac{a}{s^2 + a^2}$
$\cos at$	$\frac{s}{s^2 + a^2}$
$e^{-at} \sin bt$	$\frac{b}{(s+a)^2 + b^2}$
$e^{-at} \cos bt$	$\frac{s+a}{(s+a)^2 + b^2}$
$\frac{df}{dt}$	$sF(s) - f(0)$
$\int_0^\infty f(\tau) d\tau$	$\frac{F(s)}{s}$

Laplace transform table